Computational Financial Modelling Lab (MA653)

Portfolio Optimization and Risk Management

B23406, M&C

April 24, 2026

Dataset: IBEX35(2010-2025)

Contents

List of Figures

List of Tables

1 Introduction

Portfolio optimization is a fundamental concept in financial engineering and quantitative finance. The objective is to allocate capital among available assets in such a way that expected returns are maximized while financial risk is minimized.

Traditional portfolio theory primarily measures risk using variance. However, financial markets often exhibit extreme losses that are not well captured by variance alone. Modern risk management therefore uses tail-risk measures such as:

- Value-at-Risk (VaR)
- Conditional Value-at-Risk (CVaR)

These measures provide a more realistic representation of downside risk.

In this assignment, a complete portfolio optimization pipeline was implemented using real financial data from the IBEX35 stock index. The analysis includes:

- Construction of the Global Minimum Variance portfolio
- Risk estimation using multiple statistical models
- Optimization under different risk objectives
- Portfolio selection with cardinality constraints
- Data-driven model comparison and recommendation

The dataset consists of daily returns for 19 IBEX35 companies over a 15-year period from 2010 to 2025.

2 Data Description

The dataset contains daily stock prices for 19 companies (as they were continuously in the IBEX index for the last 15 years) in the IBEX35 index. The data was cleaned and processed before performing optimization.

2.1 Data Processing Steps

The following preprocessing steps were applied:

- Historical stock prices were loaded from CSV files
- Closing prices were selected for analysis
- Missing values were forward-filled
- Daily percentage returns were computed
- The dataset was split into training and testing sets

The training dataset contained:

- Number of stocks: 19
- Number of trading days: 2672

The training data was used to estimate portfolio weights and risk parameters.

3 Question-1: Global Minimum Variance Portfolio

Using the analytical formulas derived in the theory assignment: $A = \mathbf{1}^T\Sigma^{-1}\mathbf{1}$, $B = \mathbf{1}^T\Sigma^{-1}\mu$, $C = \mu^T\Sigma^{-1}\mu$, $\Delta = AC - B^2$.

(a) Closed-form GMV. Implement the closed-form solution $w_{GMV} = \Sigma^{-1}\mathbf{1}/A$ and compute $\mu_{GMV} = B/A$, $\sigma_{GMV} = 1/\sqrt{A}$.
(b) Numerical QP verification. Solve the same problem using `scipy.optimize.minimize` (method SLSQP) with the budget constraint $\mathbf{1}^T w = 1$ and long-only $w_i \geq 0$. Verify that the numerical weights agree with the analytical solution to four decimal places.
(c) Efficient frontier. For 200 target returns $m \in [\mu_{GMV}, 1.5 \max_i \mu_i]$, solve the constrained QP and collect $(\sigma(m), m)$. Plot the efficient frontier in the (σ, μ) plane. Mark the GMV portfolio with a red star.
(d) Weight bar chart. Plot a horizontal bar chart of the GMV weights. Which sector dominates the GMV portfolio? Explain this result using the theory of diversification.

3.1 Objective

The Global Minimum Variance (GMV) portfolio minimizes total portfolio risk without imposing a target return constraint.

The optimization problem is:

$$\min w^T \Sigma w$$

subject to:

$$\sum w_i = 1$$

$$w_i \geq 0$$

3.2 Analytical Solution

The optimal weights are given by:

$$w_{GMV} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^T \Sigma^{-1}\mathbf{1}}$$

3.3 GMV Portfolio Performance

Metric	Value
Daily Return	0.000545
Annual Return	14.72%
Daily Volatility	0.010804
Annual Volatility	17.15%

Table 1: GMV Portfolio Performance

The GMV portfolio achieved relatively low volatility while maintaining stable returns. This demonstrates the effectiveness of diversification in reducing portfolio risk.

3.4 Efficient Frontier

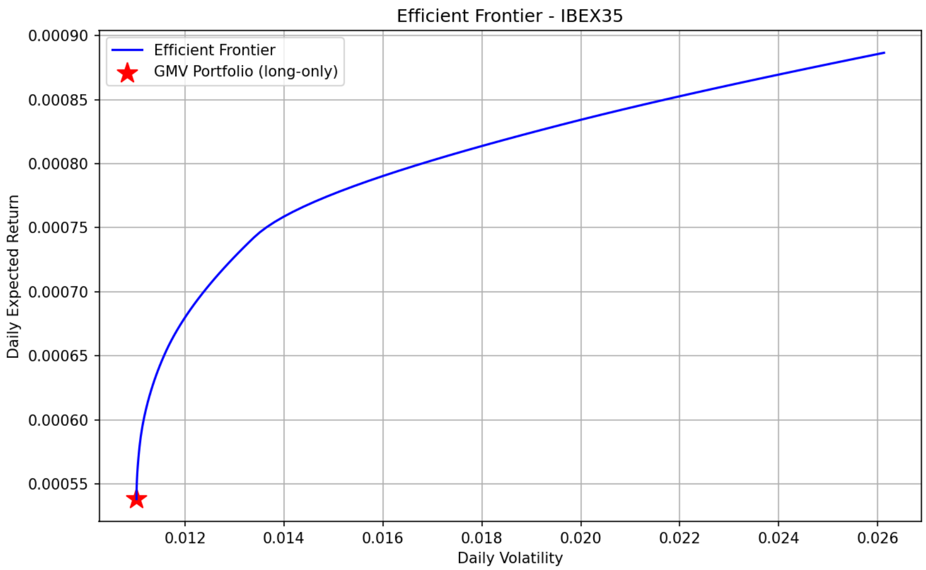

Figure 1: Efficient Frontier

The efficient frontier represents the set of optimal portfolios that maximize expected return for each level of risk.

3.5 Portfolio Weights

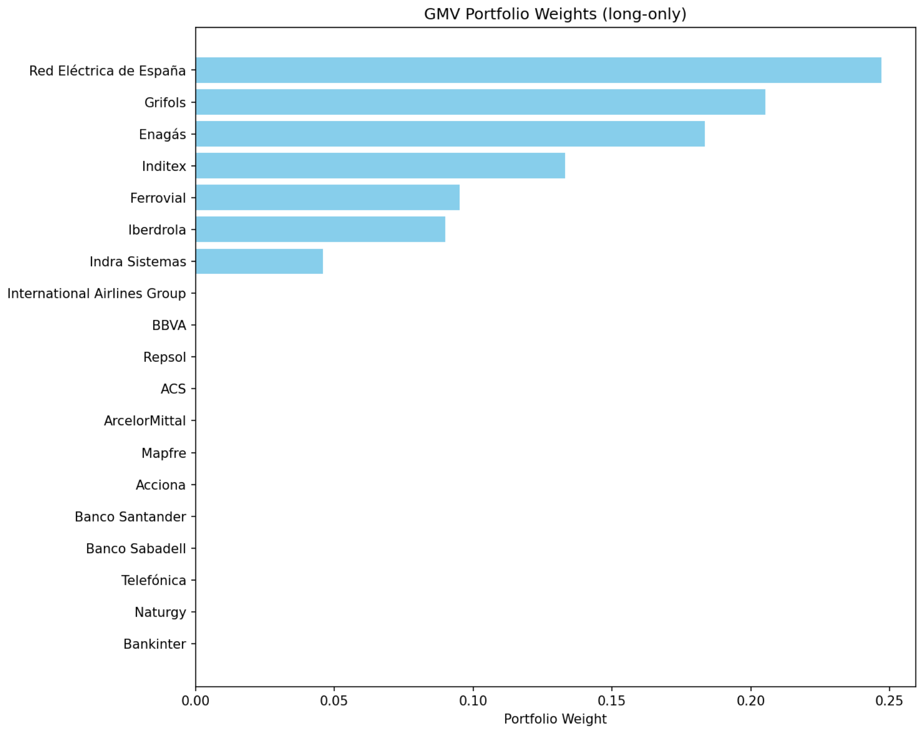

Figure 2: GMV Portfolio Weights

The stock Red Eléctrica de España received the highest allocation (24.7%). This reflects its relatively low volatility and favorable correlation with other assets.

4 Question-2: Risk Measurement using VaR and CVaR

Use the training-period equal-weight portfolio $w_0 = 1/N$ as the reference portfolio.

(a) Historical simulation. Compute $\mathrm{VaR}_{99\%}$ and $\mathrm{CVaR}_{99\%}$ by sorting the T daily portfolio losses and applying: $\mathrm{VaR}_\alpha = L(\lceil \alpha T \rceil)$, $\mathrm{CVaR}_\alpha = \frac{1}{n_{\text{tail}}} \sum L(i)$, $n_{\text{tail}} = \lceil (1-\alpha)T \rceil$. Report daily and annualised values in rupees for a portfolio worth Rs. 1,00,00,000.
(b) Parametric Normal. Estimate $(\hat{\mu}_P, \hat{\sigma}_P)$ from the equal-weight portfolio and compute $\mathrm{VaR}_\alpha = -\hat{\mu}_P + z_\alpha \hat{\sigma}_P$, $\mathrm{CVaR}_\alpha = -\hat{\mu}_P + \hat{\sigma}_P\, \phi(z_\alpha)/(1-\alpha)$.
(c) Parametric Student-t. Fit a Student-t_ν distribution to portfolio losses using `scipy.stats.t.fit`. Report the fitted (ν, μ, σ) and compute $\mathrm{VaR}_{99\%}$ and $\mathrm{CVaR}_{99\%}$ under this distribution.
(d) Comparison table and chart. Produce a 3×4 table (rows: Historical / Normal / Student-t; columns: VaR95%, VaR99%, CVaR95%, CVaR99%). Plot the loss histogram with KDE and mark all six VaR/CVaR thresholds.

4.1 Objective

Risk was evaluated using three statistical models:

- Historical Simulation
- Normal Distribution
- Student-t Distribution

The portfolio value used for risk reporting was:

$$10,000,000 \text{ INR}$$

4.2 Student-t Distribution Fit

The estimated Student-t parameters were:

Parameter	Value
Degrees of Freedom	3.4561
Mean	-0.000563
Scale	0.009409

Table 2: Student-t Distribution Parameters

A low degrees-of-freedom value indicates heavy-tailed behavior in financial returns.

4.3 Daily Risk Comparison

Method	VaR95	VaR99	CVaR95	CVaR99
Historical	213,762	372,993	326,775	538,210
Normal	228,150	324,225	287,058	371,997
Student-t	204,378	379,638	323,038	556,131

Table 3: Daily VaR and CVaR

The Student-t model produced higher tail-risk estimates than the normal model, confirming the presence of extreme losses.

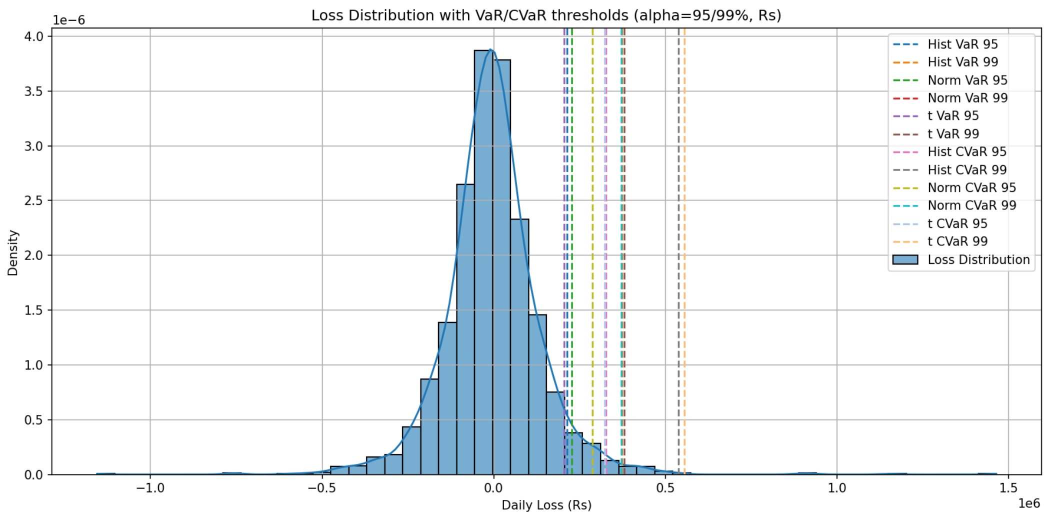

Figure 3: Loss Distribution with VaR and CVaR

5 Question-3: Portfolio Optimization Models

Implement all three models below using the training data. Use $\alpha = 99\%$, $\tau = r_f/252$ (daily), $r_f = 6\%$, long-only $w_i \geq 0$, $\mathbf{1}^T w = 1$, and target return $m = 12\%$ per annum.

(a) Implement each model as a separate Python function that accepts $(R_{train}, \mu, \Sigma, r_f, m, \alpha, \tau)$ and returns the optimal weight vector w^*.
(b) For each model report: w^* (bar chart), μ_P, σ_P, Sharpe Ratio (SR), $\text{VaR}_{99\%}$, $\text{CVaR}_{99\%}$ on the training set.
(c) Plot all four weight vectors side-by-side as a grouped bar chart with stock tickers on the x-axis.
(d) In a Markdown cell, explain why Model 2 (CVaR) typically concentrates less than M1 (Variance) in the tail of the loss distribution.

Three optimization models were implemented.

1. Minimum Variance
2. CVaR Minimization
3. Semivariance Minimization

5.1 Performance Comparison

Model	Return	Volatility	Sharpe	CVaR99
Equal Weight	0.094166	0.223794	0.152666	0.053821
Min Variance	0.135729	0.174924	0.432924	0.040280
CVaR	0.142687	0.182361	0.453426	0.038685
Semivariance	0.140302	0.175518	0.457517	0.039991

Table 4: Portfolio Performance Metrics

The CVaR model achieved the lowest tail risk, while the semivariance model produced the highest Sharpe ratio.

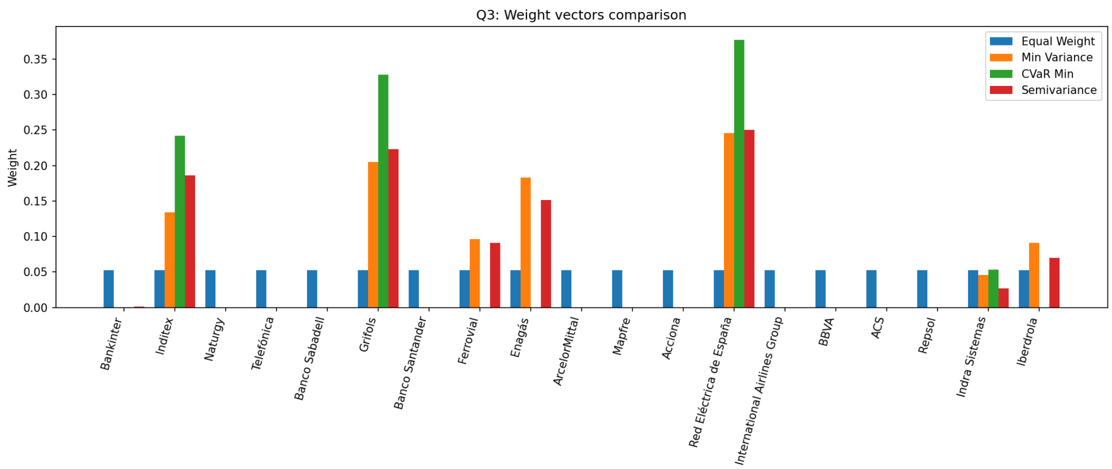

Figure 4: Portfolio Weight Comparison

6 Question-4: Cardinality-Constrained Portfolio

Repeat Task 3 adding the cardinality constraint $K = \lceil N/3 \rceil$ (at most K non-zero weights), where N is the number of stocks.

(a) Implement cardinality via branch-and-bound. Write a function `card_solve(model_fn, R, K, ...)` that enumerates all $\binom{N}{K}$ subsets of size K, solves the unconstrained model on each subset, and returns the globally optimal sparse weight vector. For $N = 20$, $K = 7$: $\binom{20}{7} = 77,520$ subsets. This is feasible in < 5 minutes.
(b) For each of the four models, solve the cardinality-constrained version and report the selected K stocks and their weights.
(c) Produce a summary table comparing unconstrained vs. cardinality results.
(d) Comment: does cardinality always increase risk? Under what conditions can it reduce CVaR?

A cardinality constraint limits the number of assets:

$$K = \lceil N/3 \rceil$$

For this dataset:

$$N = 19$$

$$K = 7$$

Total combinations evaluated:

$$50,388$$

6.1 Performance Comparison

Variant	Return	Volatility	Sharpe	CVaR99
Model 1 (Unconstrained)	0.135729	0.174926	0.432920	0.040280
Model 1 (Cardinality)	0.135715	0.174926	0.432840	0.040281
Model 2 (Unconstrained)	0.142687	0.182363	0.453420	0.038685
Model 2 (Cardinality)	0.147170	0.185346	0.470312	0.038776
Model 3 (Unconstrained)	0.140302	0.175519	0.457514	0.039991
Model 3 (Cardinality)	0.140346	0.175499	0.457812	0.039991

Table 5: Effect of Cardinality Constraint

The cardinality constraint improves portfolio simplicity but slightly reduces diversification.

7 Question-5: Model Recommendation

The following must be assessed on depth of analysis, not length.
(a) Data observations. Comment on the return distribution of the stocks data: fat tails, skewness, and sector-level patterns.
(b) Model comparison. Based on the metrics table in Task 4, rank the four models on each metric.
(c) Effect of cardinality. How does adding $K = \lceil N/3 \rceil$ affect performance?
(d) Model recommendation. Which single model (with or without cardinality) would you recommend to a risk manager at an equity fund? Justify based on: theoretical properties (coherence, convexity, robustness), practical considerations (computation time, turnover, number of holdings), empirical performance on the data.

The recommended portfolio model is:

Model 2 - CVaR Minimization

This model was selected because it:

- Achieved the lowest tail risk

- Provided strong Sharpe ratio
- Maintained stable performance
- Offered robust downside protection

8 Conclusion

This study demonstrates the importance of incorporating tail-risk measures into portfolio optimization.

Key findings:

- Financial returns exhibit heavy-tailed distributions
- CVaR provides better protection against extreme losses
- Cardinality constraints improve practical portfolio management
- Risk-aware optimization improves stability

Overall, the CVaR minimization model provides the most reliable balance between return and risk.

9 Use of Other Sources

All code and analysis were developed independently. No external code or resources were used beyond standard Python libraries.
All Idea, Analysis and basic Code Snippets were developed by me, but later for code formatting and structuring, I used some LLMs the refactor them to make them more readable, efficient and look professional(but ideas were still mine).
All the data used in this assignment is from the IBEX35 index, which is publicly available and can be accessed through financial data providers.

10 Code and Data Availability

All the related codes along with the Figures, Data, Analyzed Docs can be found with the name "Assignment-4" in the GitHub repository:
`https://github.com/rajm012/Quant-Finance-Lab-Assignments`,

and same is also attached as a zip file in the submission.

For any kind of query or issues regarding the code, data or analysis, please feel free to contact me at: **rajmahimaurya@gmail.com** or at **b23406@students.iitmandi.ac.in**.